UNIT I: Introduction to Smart Manufacturing

Subject: 23A03703d - Smart Manufacturing

1. Introduction to Smart Manufacturing

1.1 Definition

Smart manufacturing is the use of connected, data-driven and intelligent technologies to make production faster, more flexible, more efficient and more reliable. Machines, sensors, software and people share information in real time so that decisions can be made on the shop floor, not weeks later.

KEY: Smart manufacturing = production systems that sense, communicate, analyse and act automatically (with human oversight).

- It combines operational technology (OT: machines, PLCs, sensors) with information technology (IT: databases, cloud, analytics).
- Goal: right product, right quality, right time, at the lowest cost and with the least waste.

1.2 Evolution of Manufacturing

Manufacturing has moved through four major revolutions. Each one changed how goods are made.

Stage	Period (approx.)	Key idea	Main technology
Industry 1.0	late 1700s - 1800s	Mechanisation	Water and steam power, mechanical looms
Industry 2.0	late 1800s - early 1900s	Mass production	Electricity, assembly lines
Industry 3.0	1970s - 2000s	Automation	PLCs, robots, computers, ERP
Industry 4.0	2011 onwards	Smart and connected	IoT, CPS, cloud, AI, big data

- The term "Industry 4.0" was introduced by the German government's high-tech strategy initiative in 2011.
- Industry 5.0 (promoted by the European Union from about 2021) adds human-centric design, sustainability and resilience.

1.3 Importance of Smart Manufacturing

- Higher productivity: machines run longer and work better together.
- Better quality: defects are detected early by sensors and vision systems.
- Lower cost: less energy, less scrap and less downtime.
- Flexibility: lines can switch to new products quickly (mass customisation).
- Faster decisions: dashboards and analytics show problems as they happen.
- Safer and more sustainable working conditions.

2. Industry 4.0 and Its Impact on Manufacturing

2.1 Nine pillars of Industry 4.0

- Industrial Internet of Things (IIoT)
- Cyber-physical systems (CPS)
- Cloud and edge computing
- Big data and analytics
- Artificial intelligence and machine learning
- Digital twin and simulation
- Additive manufacturing (3D printing)
- Augmented and virtual reality (AR/VR)
- Cybersecurity

2.2 Impact on manufacturing

- Products: smarter products that report their own condition and usage.
- Processes: self-optimising production lines and predictive maintenance.
- Business models: "product as a service", where customers pay for uptime or output.
- Supply chains: end-to-end visibility from supplier to customer.
- Workforce: fewer routine tasks; more demand for data, software and maintenance skills.

KEY: Key idea: Industry 4.0 connects the physical factory to the digital world, so the factory can sense itself and improve itself.

3. Traditional vs Smart Manufacturing

Aspect	Traditional manufacturing	Smart manufacturing
Data	Collected manually, often on paper	Collected automatically from sensors
Maintenance	Reactive or fixed schedule	Predictive, based on machine condition
Quality control	Sample inspection after production	Inline, 100% monitoring with vision and AI
Production planning	Static plans, updated rarely	Dynamic, updated from live data
Decision making	Based on experience and reports	Based on real-time analytics
Flexibility	Hard to change products	Quick reconfiguration of lines
Communication	Hierarchical, slow	Connected, machine to machine

- Traditional plants are not "bad", but they react to problems; smart plants anticipate them.
- Upgrading is usually gradual: add sensors and data first, then analytics, then automation.

4. Key Technologies in Smart Manufacturing

- Sensors and actuators: measure temperature, pressure, vibration, position, and act on the process.
- Industrial networks: fieldbus and industrial Ethernet links between devices.
- IIoT platforms: collect, store and visualise machine data.
- Edge computing: process data close to the machine for speed.
- Cloud computing: scalable storage and analytics across many plants.
- Big data analytics: finds patterns in very large data sets.
- AI and machine learning: predicts failures, classifies defects, optimises settings.
- Digital twin: a virtual copy of a machine, line or plant.
- Robotics and cobots: flexible automation that can work near people.
- Additive manufacturing: builds parts layer by layer from a digital model.
- Cybersecurity: protects machines and data from attacks.

5. Cyber-Physical Systems (CPS)

5.1 Basics

A cyber-physical system tightly links a physical process with computation and communication. Sensors read the physical world, software decides what to do, and actuators change the physical world. The loop runs continuously.

KEY: CPS loop: SENSE -> COMMUNICATE -> COMPUTE/DECIDE -> ACT -> SENSE again.

5.2 Components of a CPS

- Physical plant: the machine, conveyor, robot or process being controlled.
- Sensors: measure the state (for example, temperature or speed).
- Embedded controllers: local processors that run control logic.
- Communication network: wired or wireless links.
- Cyber layer: analytics, models and decision software, often in the cloud.
- Human-machine interface (HMI): screens where operators see and steer the system.

5.3 Example

A milling machine has a vibration sensor. When vibration rises above a threshold, the controller slows the spindle and sends an alert. The maintenance team sees the event on a dashboard and schedules a tool change before the part is damaged.

Exam Q: Explain the CPS loop with an example from a factory.

6. Internet of Things (IoT) in Manufacturing

6.1 What is IoT?

IoT means everyday physical objects are given sensors, a network connection and an identity, so they can send and receive data. In manufacturing this is called the Industrial Internet of Things (IIoT).

6.2 Layers of an IIoT architecture

- Device layer: sensors, actuators, drives, PLCs, robots.
- Connectivity layer: fieldbus, Ethernet, Wi-Fi, 5G, LoRaWAN.
- Edge layer: gateways that filter, aggregate and secure data.
- Platform layer: cloud or on-premise storage, analytics and device management.

- Application layer: dashboards, MES, maintenance apps, mobile alerts.

6.3 Uses of IoT on the shop floor

- Asset tracking and machine status monitoring (OEE dashboards).
- Energy monitoring of motors, compressors and furnaces.
- Remote monitoring of machines in several plants.
- Track-and-trace of parts and batches for quality records.
- Environmental monitoring of temperature and humidity in clean rooms.

7. Smart Sensors and Actuators

7.1 Smart sensors

A smart sensor is a sensor with built-in processing. Besides measuring, it can calibrate itself, check its own health, filter noise and send data in a standard format.

- Types: temperature (thermocouple, RTD), proximity (inductive, capacitive), vision cameras, vibration (accelerometer), flow meters, pressure transmitters.
- Advantages: less wiring, self-diagnosis, plug-and-play integration.

7.2 Actuators

An actuator converts a control signal into physical motion or action.

- Electric: motors, servo drives, stepper motors.
- Pneumatic: air cylinders and valves.
- Hydraulic: heavy-duty presses and lifts.
- Solenoid valves and relays for on/off switching.

KEY: Sensor measures, controller decides, actuator acts.

8. Communication Protocols

A protocol is a set of rules that devices follow to exchange data. Smart factories use several protocols because each suits a different job.

8.1 Modbus

- Created in 1979 by Modicon (now part of Schneider Electric) for PLC communication.
- Simple master/slave (client/server) model; the master polls the slaves.
- Two common versions: Modbus RTU (serial, RS-485) and Modbus TCP (over Ethernet, usually port 502).
- Strength: very simple and supported by almost every device. Weakness: no built-in security and limited data model.
- Typical use: reading meters, drives and simple I/O.

8.2 MQTT (Message Queuing Telemetry Transport)

- A lightweight publish/subscribe protocol designed for unreliable or low-bandwidth networks.
- Devices publish messages to a topic; a broker distributes them to subscribers.
- Very small packet overhead; supports quality-of-service levels 0, 1 and 2.
- Standard port 1883 (8883 with TLS).
- Typical use: sending sensor data from many devices to a cloud or edge platform.

KEY: MQTT is the "postal service" of IoT: senders publish to a topic, and any interested receiver subscribes to it.

8.3 OPC-UA (OPC Unified Architecture)

- Standard IEC 62541, platform-independent and vendor-neutral.
- Provides a rich information model: data is described with meaning, units and relationships.
- Includes built-in security (encryption, certificates, user authentication).
- Works across Windows, Linux and embedded devices, and across vendors.
- Default port 4840.

- Typical use: connecting PLCs, SCADA, MES and cloud systems in a secure, structured way.

8.4 Comparison

Protocol	Model	Security	Best for
Modbus	Master/slave, request-response	Weak (none built in)	Simple device reading
MQTT	Publish/subscribe via broker	TLS and login	IoT telemetry, cloud links
OPC-UA	Client/server with rich model	Strong, built in	Secure industrial interoperability

8.5 Worked example: how the three protocols work togeth

A CNC machine in a plant has a spindle temperature sensor, a PLC and a cloud dashboard.

- The PLC reads the sensor over Modbus TCP (simple register reads, fast local loop).
- An edge gateway receives the register values through the PLC's OPC-UA server and adds meaning: the value is named "SpindleTemp", its unit is degrees Celsius and its limit is 80.
- The gateway publishes readings with MQTT to topic "plant1/line3/cnc2/temperature" every 5 seconds.
- The cloud dashboard subscribes to that topic and draws the trend graph; an alert service subscribes to the same topic and sends an SMS if the temperature stays above the limit.

KEY: Modbus gets the data out of the device, OPC-UA describes and secures it, and MQTT moves it efficiently to other systems.

4.1 Benefits and challenges of smart manufacturing

Benefits

- Reduced unplanned downtime through predictive maintenance.
- Lower scrap and rework through real-time quality checks.
- Better use of energy and raw material.
- Shorter time-to-market for new products.
- Better traceability for audits and recalls.

Challenges

- High initial investment in sensors, networks and software.
- Integration of old machines (legacy equipment) that have no digital interfaces.
- Shortage of skilled staff who know both shop-floor and data skills.
- Data overload: collecting data is easy, using it well is hard.
- Cybersecurity risks when machines are connected to networks.
- Interoperability between devices from different vendors.

KEY: Most successful plants start with a clear business problem (for example, reducing machine breakdowns) rather than buying technology for its own sake.

5.4 The 5C architecture for cyber-physical systems

Researchers (Lee, Bagheri and Kao, 2015) proposed a 5-level architecture to build CPS step by step:

- Level 1 - Connection: acquire reliable data from machines and sensors.
- Level 2 - Conversion: turn data into information, for example by detecting degradation trends.
- Level 3 - Cyber: connect machines to each other and to a central analysis, forming a digital model of the fleet.

- Level 4 - Cognition: present knowledge to experts so that they can make decisions.
- Level 5 - Configuration: feed decisions back to machines for self-adjustment (supervisory control).

Level	Question it answers	Example
Connection	Can we get the data?	Vibration sensor on a motor
Conversion	What does the data mean?	Vibration trend shows bearing wear
Cyber	How does this machine compare with others?	Compare bearing wear across 20 motors
Cognition	What should we do?	Recommend bearing replacement next week
Configuration	How should the system react?	Reduce speed until replacement

5.5 Real-time constraints in CPS

- Many CPS control loops must respond in milliseconds; a delay can damage equipment.
- Hard real-time systems guarantee a deadline; soft real-time systems tolerate occasional delay.
- Time synchronisation (for example IEEE 1588 Precision Time Protocol) keeps the clocks of many devices aligned so that events can be ordered correctly.
- Edge computing is used when the cloud is too slow for the control loop.

6.4 Typical IIoT data flow

1. A sensor measures a quantity every few milliseconds.
2. The PLC or the sensor's own processor scales and checks the value.
3. The gateway aggregates data (for example, one average per second).
4. Data is sent to the platform where it is stored with a timestamp and a device identity.
5. Analytics compare the value with models or thresholds.
6. Results appear on dashboards and trigger alarms or work orders in the MES or CMMS.

7.3 Selecting a sensor (checklist)

- Measuring range and accuracy needed.
- Operating temperature, dust, water and vibration in the environment (IP rating).
- Response time and sampling rate.
- Output signal: analog (4-20 mA, 0-10 V) or digital (IO-Link, Modbus, Ethernet).
- Calibration needs and cost over the life of the system.

8.6 Quick comparison of other common industrial protoco

- PROFINET and EtherNet/IP: industrial Ethernet used for fast device communication, mostly in automation of machines.
- PROFIBUS: fieldbus used for older plants; being replaced by PROFINET.
- HART: sends digital data over analog 4-20 mA wires, common for process instruments.
- DNP3 and IEC 61850: used in utilities and power systems.
- Note: exams usually focus on Modbus, MQTT and OPC-UA, so learn their features well.

9.1 Revision checklist

- Can you define smart manufacturing in two sentences?
- Can you name the four industrial revolutions and their key technologies?
- Can you draw the CPS sense-decide-act loop?
- Can you list the layers of IIoT architecture?
- Can you explain Modbus, MQTT and OPC-UA with one advantage each?
- Can you describe at least two benefits and two challenges?

9. Summary

- Smart manufacturing connects machines, data and people to make production adaptive.
- Industry 4.0 is the fourth industrial revolution, built on CPS, IIoT, cloud, AI and digital twins.
- CPS link the physical process with computation through a sense-decide-act loop.
- Smart sensors and actuators are the eyes and hands of the system.
- Modbus, MQTT and OPC-UA carry the data, each with its own strengths.

Exam Q: List three differences between traditional and smart manufacturing.

Exam Q: What are the layers of an IIoT architecture?

Exam Q: When would you choose MQTT instead of OPC-UA?

UNIT II: Data Analytics, AI and Automation in Manufacturing

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1. Role of Big Data in Manufacturing

1.1 What is big data?

Big data refers to data sets that are too large, too fast or too varied for traditional tools (such as a single spreadsheet or a simple database) to handle. In a factory, data comes from machines, PLCs, sensors, cameras, MES, ERP and quality systems.

1.2 The 5 V's of big data

- Volume: the amount of data (terabytes of sensor logs per month).
- Velocity: the speed at which data arrives (a vibration sensor may send thousands of samples per second).
- Variety: many formats (numbers, text, images, video, alarms).
- Veracity: how accurate and trustworthy the data is.
- Value: the useful insight that can be extracted and turned into money or savings.

KEY: Data by itself is not valuable; it becomes valuable only when it leads to a better decision.

1.3 Uses of big data in manufacturing

- Finding the root cause of defects by comparing thousands of production runs.
- Optimising machine settings (speed, temperature, pressure) for the best yield.
- Forecasting demand and planning inventory.
- Tracking energy use per product.
- Improving supplier quality through shared data.

1.4 Data architecture

- Data sources: machines, sensors, MES, ERP, quality and maintenance records.
- Data ingestion: collecting data through MQTT, OPC-UA or APIs.
- Data storage: data lakes (raw data), data warehouses (structured data), time-series databases (sensor readings).
- Processing: batch processing (for historical reports) and stream processing (for live alerts) with tools such as Apache Spark or Kafka.
- Analytics: statistics, machine learning and visualisation dashboards.

1.5 Data quality issues

- Missing values when a sensor fails or a network drops.
- Noise and outliers from electrical interference.
- Inconsistent timestamps between machines (clock drift).
- Labels that are wrong or missing (for example, defects not recorded correctly).

KEY: Clean data is the foundation: a model trained on poor data will give poor results.

Exam Q: Explain the 5 V's of big data with a manufacturing example for each.

2. Machine Learning and Deep Learning for Smart Manufa

2.1 Machine learning (ML)

Machine learning lets a computer learn patterns from data instead of following fixed rules written by a programmer.

- Supervised learning: learns from labelled examples. Regression predicts a number (for example, the remaining life of a bearing). Classification predicts a category (for example, good part or defective part).
- Unsupervised learning: finds structure without labels. Clustering groups similar production batches or machine behaviours.

- Reinforcement learning: learns by trial and reward. Used to tune robot control or scheduling policies.

2.2 Common ML algorithms in factories

- Linear and logistic regression: simple, explainable baselines.
- Decision trees and random forests: good for mixed data and for showing which factors matter.
- Support vector machines: classification with small to medium data sets.
- k-means clustering: grouping operating modes of a machine.
- Gradient boosting: strong performance on tabular process data.

2.3 Deep learning (DL)

Deep learning uses neural networks with many layers. It works very well on images, sound and sequences.

- Convolutional neural networks (CNN): inspect camera images to detect scratches, cracks or missing components.
- Recurrent networks and LSTMs: model time-series sensor data, such as temperature or current over time.
- Autoencoders: learn to rebuild normal data; large rebuild error signals an anomaly.
- Transformers: increasingly used for log analysis and combining several data types.

2.4 The ML workflow

1. Define the business problem (for example, reduce motor failures by 30 percent).
2. Collect and label data.
3. Clean and engineer features (for example, average vibration per minute, peak current).
4. Split data into training, validation and test sets.
5. Train and tune models.
6. Evaluate using suitable metrics (accuracy, precision, recall, mean absolute error).
7. Deploy to the edge or cloud and monitor for drift.

8. Retrain when the process changes.

2.5 Metrics to remember

- Precision: of the items flagged as defective, how many really are.
- Recall: of all real defects, how many were flagged.
- False alarms and missed failures have different costs; choose the metric that matches the cost.

Exam Q: Why is recall more important than precision for detecting a critical equipment failure?

3. Predictive Maintenance and Anomaly Detection

3.1 Maintenance strategies

Strategy	When work is done	Drawback
Reactive (run to failure)	After breakdown	Long downtime, damage to other parts
Preventive	At fixed intervals	Parts replaced even when still good
Condition-based	When a measured condition crosses a limit	Needs sensors and thresholds
Predictive	Before failure, based on forecast	Needs data, models and skill

- Predictive maintenance uses models to estimate the remaining useful life (RUL) of a component and schedule work at the best time.

3.2 Common condition indicators

- Vibration: rising amplitude or new frequency peaks indicate bearing or gear wear.
- Temperature: a slow rise can show lubrication problems.
- Motor current signature: changes reveal rotor or load problems.
- Acoustic emission: detects cracks and leaks.

- Oil analysis: metal particles show wear.

3.3 Anomaly detection

An anomaly is a pattern that does not match normal behaviour. Methods include:

- Statistical limits: flag values beyond mean plus or minus three standard deviations.
- Moving averages and control charts (Shewhart, CUSUM) for slow drifts.
- Isolation forest: isolates unusual points quickly in multi-sensor data.
- One-class SVM: learns the boundary of normal data.
- Autoencoder reconstruction error: a high error means the pattern is unfamiliar.

KEY: Anomaly detection tells you that something is unusual; diagnosis tells you what is wrong.

3.4 Workflow of a predictive maintenance system

- Collect vibration, temperature and current data at the edge.
- Extract features in the time and frequency domains (RMS, kurtosis, FFT peaks).
- Score each window with the model; compare with thresholds.
- Raise an alert and create a work order in the CMMS.
- Record the outcome to improve the model later.

3.5 Benefits

- Less unplanned downtime and fewer emergency repairs.
- Lower spare-part inventory because parts are ordered in time.
- Longer machine life and better safety.

4. Digital Twin Technology

4.1 Definition

A digital twin is a virtual representation of a physical asset, process or system that is kept up to date with live data from the real one. It can be used to monitor, simulate, predict and optimise.

KEY: The three parts of a digital twin: the physical object, its virtual model, and the two-way data connection between them.

4.2 Types of digital twin

- Component twin: a bearing or a motor.
- Asset twin: a complete machine such as a CNC or a robot.
- System twin: a production line or a cell.
- Process or plant twin: the whole factory, including material flow and energy.

4.3 Levels of twin maturity

- Digital model: no automatic data link; a static design model.
- Digital shadow: one-way data flow from physical to virtual.
- Digital twin: two-way link; the virtual model can also send commands or changes back.

4.4 What a digital twin enables

- Virtual commissioning: test PLC programs against the virtual line before the real one is built.
- What-if analysis: check the effect of a faster speed without risking real equipment.
- Remote monitoring and diagnosis.
- Predictive maintenance using the twin's physics-based model together with data-based ML.
- Training operators in a safe virtual environment.

4.5 Building blocks

- Geometric model: CAD and 3D shape.
- Physics model: equations for heat, flow, stress and motion.
- Behaviour model: rules or learned models of how the system responds.
- Data integration: IIoT connectivity and time-series storage.
- Visualisation: 3D dashboards, AR overlays.

4.6 Example

A pump has a digital twin built from its CAD model and fluid equations. Live pressure and flow data update the twin. When the twin's predicted efficiency drops below the measured efficiency, the team inspects the impeller for wear before a leak occurs.

Exam Q: Differentiate between a digital model, a digital shadow and a digital twin.

5. Industrial Automation: PLCs, SCADA and DCS

5.1 Programmable Logic Controller (PLC)

A PLC is a rugged industrial computer that reads inputs, runs a control program and drives outputs in a repeating cycle called the scan cycle.

- Scan cycle: read inputs -> execute program -> update outputs -> handle communication -> repeat.
- Typical scan times are a few milliseconds.
- Programming standard IEC 61131-3 includes ladder diagram (LD), function block diagram (FBD), structured text (ST), sequential function chart (SFC) and instruction list.
- Used for discrete manufacturing: conveyors, packaging, assembly stations, machine tools.

5.2 Ladder logic example

A motor start-stop circuit: a START contact and a STOP contact control the motor coil. Pressing START latches the motor on through a holding contact; pressing STOP breaks the latch. In ladder notation, rungs are read left to right like rungs of a ladder, from power rail to power rail.

5.3 SCADA (Supervisory Control and Data Acquisition)

SCADA is a system that monitors and controls geographically spread or large processes from a central place.

- Components: sensors and PLCs or RTUs (remote terminal units) at the field, communication network, SCADA server, historian database, and HMI screens.
- Functions: data acquisition, supervisory control (setpoints, start and stop), alarms, trends, reports and logs.
- Used in water treatment, power distribution, oil pipelines, railways and large plants.

5.4 DCS (Distributed Control System)

A DCS is a control system in which controllers are distributed across the plant but managed in an integrated way.

- Designed mainly for continuous and batch process industries: chemicals, refineries, pharmaceuticals, power plants.
- Strong points: built-in redundancy, process control blocks (PID loops), tight integration of control and operations.
- Distinction: SCADA is strong for wide-area monitoring; DCS is strong for one process site with complex control loops.

Feature	PLC	SCADA	DCS
Main role	Local logic control	Supervisory monitoring	Integrated process control
Typical industry	Discrete manufacturing	Utilities, pipelines	Chemicals, refineries
Scale	Machine or cell	Wide area	Single large plant
Redundancy	Optional	Server level	Built in

Exam Q: Compare PLC, SCADA and DCS in a short table.

6. Autonomous Robots and Cobots

6.1 Industrial robots

An industrial robot is a programmable mechanical arm with several joints (axes). Typical types include articulated, SCARA, delta and Cartesian robots. They are used for welding, painting, pick-and-place, machine tending and assembly.

6.2 Autonomous mobile robots (AMRs) and AGVs

- Automated guided vehicles (AGVs) follow fixed paths such as magnetic tape or wires.
- Autonomous mobile robots (AMRs) use sensors, maps and planning to find their own route around people and obstacles.
- Used for moving materials between stations and warehouses.

6.3 Collaborative robots (cobots)

A cobot is designed to work safely alongside people without a safety cage.

- Features: force and torque sensing, rounded joints, speed and separation monitoring, power and force limiting.
- Safety standards: ISO 10218 for industrial robots and ISO/TS 15066 for collaborative operation.
- Advantages: easy to program by hand-guiding, quick to redeploy, low cost for small batches.
- Limitations: slower and lower payload than traditional industrial robots.

KEY: Cobots do not replace safety analysis; each application must still be risk-assessed.

6.4 Robot programming methods

- Teach pendant: operator moves the robot and records points.
- Hand guiding: operator moves the arm physically (common for cobots).
- Offline programming: program built in simulation and then loaded.
- Vision-guided picking: camera finds the part position and robot adapts.

Exam Q: State two differences between an AGV and an AMR.

7. Additive Manufacturing (3D Printing)

7.1 Basic idea

Additive manufacturing builds a part layer by layer from a digital 3D model, adding material only where it is needed. It is the opposite of subtractive machining, which removes material.

7.2 Main processes

- Fused deposition modelling (FDM / material extrusion): melted thermoplastic is extruded through a nozzle.
- Stereolithography (SLA / vat photopolymerisation): a UV laser cures liquid resin layer by layer.
- Selective laser sintering (SLS / powder bed fusion): a laser fuses nylon or other powder.
- Metal powder bed fusion (SLM, DMLS): a laser melts metal powder for functional metal parts.
- Binder jetting: a liquid binder joins powder layers; parts are then sintered.
- Directed energy deposition: metal is melted as it is deposited, used for repairs.

7.3 Advantages in smart manufacturing

- Complex geometry at no extra cost (internal channels, lattices).
- Small batches and customised products.
- Rapid prototyping and fewer tooling costs.
- Spare parts printed on demand near the point of use.
- Lightweight parts for aerospace and automotive.

7.4 Limitations

- Slower than mass moulding for large volumes.
- Surface finish and dimensional accuracy may need post-processing.
- Material properties and repeatability need careful control and inspection.

7.5 Digital thread

A digital thread connects the design file, print parameters, build records and inspection data for each part. This allows traceability and quality certification.

8. Case Studies of Smart Factories

8.1 Siemens Electronics Works Amberg (Germany)

- A highly automated plant producing automation controllers. Products are tracked through the line, and the plant is often cited as a reference for digital manufacturing.
- Features: extensive use of PLCs, MES, robotics and digital planning, with very high quality levels reported.
- Lessons: integrate IT and OT early; use the data from one stage to improve the next.

8.2 BMW Spartanburg and other automotive plants

- Automotive plants use robots, vision systems and AGVs/AMRs for flexible assembly.
- Lessons: flexible lines that can build several models; digital planning with simulation before construction.

8.3 General lessons from smart factory cases

- Start with a clear business goal such as less downtime or less scrap.
- Pilot one line, measure results, then scale to more lines.
- Invest in skills and change management as much as in technology.
- Measure benefits with agreed KPIs such as OEE, first-pass yield and energy per unit.

KEY: OEE (Overall Equipment Effectiveness) = Availability x Performance x Quality.

Exam Q: Write the steps you would follow to plan a predictive maintenance project in a plant.

9. Worked Examples and Revision

9.1 Worked example: bearing health monitoring

A motor bearing is monitored with an accelerometer sampling at 10 kHz.

- Step 1: every minute, compute the RMS vibration value of the last 60 seconds.
- Step 2: learn the normal range from six weeks of healthy data (for example, 0.8 to 1.2 mm/s).
- Step 3: set an alert at 1.8 mm/s and a shutdown limit at 2.5 mm/s.
- Step 4: use an autoencoder on frequency-domain features to catch new fault types that thresholds miss.
- Step 5: when the alert fires, the CMMS creates a work order; the technician replaces the bearing in the next planned stop.
- Result: the failure is caught before the motor overheats, avoiding a two-day line stop.

KEY: Always combine simple thresholds (fast and explainable) with advanced models (sensitive to new patterns).

9.2 Worked example: choosing a robot for a packaging cell

- Task: pick cartons weighing 3 kg and place them on a pallet, working next to a packer.
- Requirement: human workers enter the cell for refilling, so the robot must be a cobot.
- Check payload (at least 3 kg plus gripper weight), reach, speed and safety features.
- Decision: a cobot with force limiting and a safety-rated stop, programmed by hand guiding.

9.3 Worked example: PLC logic for a conveyor

- Inputs: START button, STOP button, photo sensor at the end of the belt.
- Output: conveyor motor.

- Logic: motor runs when START is pressed and STOP is not pressed; the motor stops automatically when the photo sensor sees a full box for 3 seconds.
- Test: simulate each input in the PLC software before connecting the motor.

9.4 Revision checklist

- Can you explain the 5 V's of big data?
- Can you name a supervised and an unsupervised learning method with an example?
- Can you describe the difference between anomaly detection and diagnosis?
- Can you draw the PLC scan cycle?
- Can you compare SCADA and DCS?
- Can you list two additive manufacturing processes and their uses?
- Can you define OEE?

9.5 Short answer practice

Exam Q: What is the role of an edge gateway in a predictive maintenance system?

Exam Q: Why do cobots need force and torque sensing?

Exam Q: Give two reasons why a digital twin is more useful than a static CAD model.

UNIT III: Cloud, Edge Computing, RFID and Blockchain in Manufacturing

Subject: 23A03703d - Smart Manufacturing

1. Introduction to Cloud Computing for Manufacturing

1.1 What is cloud computing?

Cloud computing means renting computing resources (servers, storage, databases, analytics and software) over the internet instead of owning them. The provider manages the hardware, and the user pays for what is used.

1.2 Service models

- IaaS (Infrastructure as a Service): virtual machines and storage. The user manages the operating system and applications.
- PaaS (Platform as a Service): a ready development and runtime platform, for example for building IoT applications.
- SaaS (Software as a Service): finished applications used through a browser, for example a cloud MES or ERP.

1.3 Deployment models

- Public cloud: shared infrastructure run by a provider (AWS, Microsoft Azure, Google Cloud).
- Private cloud: dedicated to one company, often in its own data centre.
- Hybrid cloud: a mix; sensitive or real-time work stays local, while analytics and archives go to the public cloud.
- Community cloud: shared by organisations with common needs, for example a regional industry group.

1.4 Benefits for manufacturing

- Scalability: add capacity for peak analysis without buying servers.
- Lower upfront cost: no large investment in data centres.

- Access from anywhere: managers and suppliers can see live data securely.
- Collaboration: plants, suppliers and customers share one data platform.
- Fast updates and access to advanced tools such as machine learning services.

1.5 Limitations

- Latency: round trips to a distant cloud can be too slow for control loops.
- Connectivity: a plant that loses its internet link may lose visibility.
- Data sovereignty and confidentiality: rules may limit where data can be stored.
- Recurring cost and vendor lock-in.

KEY: Cloud is excellent for analytics, planning and long-term storage; it is not a substitute for fast local control.

Exam Q: Give two reasons a plant might prefer a hybrid cloud model.

2. Edge and Fog Computing in Real-time Manufacturing Sy

2.1 Edge computing

Edge computing processes data close to where it is produced, on gateways, industrial PCs or devices themselves. Only useful results or summaries are sent to the cloud.

- Benefits: very low latency, works during internet outages, reduces bandwidth costs, keeps sensitive raw data on site.
- Examples: a vision system that rejects a defective bottle within 20 milliseconds; a gateway that averages 10,000 vibration samples into one value per second.

2.2 Fog computing

Fog computing is a layer between the edge and the cloud. It spreads computing across routers, local servers and gateways in the network so that data can be processed at intermediate points.

- Edge: on the machine or in the device.
- Fog: in the plant network, for example a local server that coordinates several cells.
- Cloud: centralised, for big data storage and enterprise analytics.

Layer	Location	Typical latency	Typical task
Device/edge	On or next to machine	Milliseconds	Control, filtering, immediate alarms
Fog	Plant server or network node	Tens of milliseconds	Line coordination, local dashboards
Cloud	Remote data centre	Hundreds of milliseconds or more	Analytics, ML training, enterprise reports

2.3 Real-time requirements

- Hard real-time: missing a deadline is unacceptable (for example, a safety stop).
- Soft real-time: occasional delay is tolerable (for example, a dashboard refresh).
- Edge and fog computing make hard real-time closed loops possible while still using cloud intelligence.

2.4 Architecture example

A bottling line: PLCs run the filling valves (device layer); an edge gateway checks fill level and sends commands if the level drifts (edge); a fog server coordinates the capper and labeller (fog); the cloud stores batch records and runs yield analysis every night (cloud).

Exam Q: Explain why a safety stop should not depend on a cloud connection.

3. Cloud-based Manufacturing Execution Systems (MES)

3.1 What is an MES?

A Manufacturing Execution System controls and tracks production on the shop floor. It sits between the planning system (ERP) and the machines (PLCs, SCADA).

- Functions: work order management, production tracking, resource allocation, quality management, material traceability, labour tracking, performance analysis and maintenance coordination.

3.2 Where MES fits in the automation pyramid

- Level 4: ERP (business planning, finance, orders).
- Level 3: MES (production execution and scheduling).
- Level 2: SCADA and HMI (supervision of processes).
- Level 1: PLCs and drives (control).
- Level 0: sensors and actuators (field devices).

3.3 Cloud-based MES

A cloud MES is delivered as a service (SaaS). Plants access it through web browsers and connect machines through an edge gateway.

- Advantages: quick deployment, automatic updates, multi-plant views, lower IT burden, pay-per-use pricing.
- Concerns: data security, dependence on the internet, integration with legacy systems, and the need for local fallback.
- Hybrid designs keep critical execution local and use the cloud for analytics and cross-site comparison.

3.4 Key KPIs an MES tracks

- OEE (availability x performance x quality).
- First-pass yield and scrap rate.
- Cycle time and throughput.
- Work-in-progress (WIP) levels.
- On-time delivery.

3.5 Example: traceability in pharmaceuticals

Each batch receives a unique ID. The MES records every operation, operator, machine setting, material lot and test result. If a problem is found later, the batch history can be traced in minutes instead of days.

KEY: MES connects "what should be made" (ERP) with "what is being made" (machines).

Exam Q: Name three functions of an MES and one benefit of moving it to the cloud.

4. Smart Sensors and RFID in Inventory Management

4.1 Smart sensors in inventory

Smart sensors on shelves, bins, pallets and vehicles track quantity, location, temperature, humidity and shock. They can alert staff when a bin is nearly empty or when cold-chain conditions are broken.

4.2 What is RFID?

Radio Frequency Identification (RFID) uses radio waves to read and write data on a tag attached to an object, without line of sight. A reader sends a signal; the tag responds with its unique ID.

4.3 Components of an RFID system

- Tag: a microchip with an antenna. Passive tags have no battery and draw power from the reader signal; active tags have their own battery and a longer range.
- Reader (interrogator): transmits and receives radio signals.
- Antenna: part of the reader that sends and receives.
- Middleware and database: filters the reads and stores them with time and location.

4.4 RFID frequency bands

- Low frequency (LF, about 125 to 134 kHz): short range, good near metal or liquids; used for animal tags and access control.

- High frequency (HF, 13.56 MHz): medium range; used for library books and payment cards (NFC is a form of HF RFID).
- Ultra high frequency (UHF, about 860 to 960 MHz): long range, fast bulk reading; used for pallets and supply chains.
- Microwave (2.45 GHz and above): very long range, used for toll and vehicle tracking.

4.5 Uses in inventory management

- Automatic stock counting without opening boxes.
- Goods receipt and dispatch with dock-door readers.
- Work-in-progress tracking on the shop floor (tagged pallets or carriers).
- Tool management: checking tools in and out of a crib.
- Reduced stock-outs and less manual error.

4.6 Advantages and limitations of RFID

- Advantages: no line of sight needed, many tags read at once, reusable tags, rich data.
- Limitations: metal and liquids can disturb signals; tag cost; read collisions when many tags are close; privacy concerns when tags are used on products sold to consumers.

KEY: RFID tells you WHICH item is where; barcodes need a clear line of sight and are read one at a time.

Exam Q: Compare a barcode and an RFID tag for tracking pallets in a warehouse.

5. Blockchain in Manufacturing

5.1 What is blockchain?

Blockchain is a distributed, append-only ledger. Records are grouped into blocks; each block contains a cryptographic hash of the previous block, so changing an old record breaks the chain. Many parties hold a copy and agree on new entries through a consensus mechanism.

5.2 Key ideas

- Distributed: no single owner controls the record.
- Immutable: past entries are very hard to change without detection.
- Transparent: all permitted parties see the same history.
- Consensus: entries are validated by agreed rules (for example, proof of authority in private networks).

5.3 Types of blockchain

- Public: open to anyone (for example Bitcoin, Ethereum).
- Private: run by one organisation.
- Consortium (permissioned): run by a group of known companies; common in supply chains.

5.4 Blockchain for secure supply chains

- Traceability: record each step from raw material to final product (origin, processing, shipping).
- Authenticity: prove that parts are genuine and not counterfeit.
- Compliance: store certificates, test results and audit trails that cannot be quietly edited.
- Trust between suppliers: parties who do not trust each other can share the same record.
- Payment and claims: automatic settlement when goods arrive in good condition.

5.5 Example

A food producer records harvest date, farm, temperature history from IoT sensors and certification on a consortium blockchain. Retailers scan a code and see the complete history. If a contamination is found, the affected batches can be identified within minutes.

5.6 Limitations

- Scalability and speed: public chains can be slow and costly.
- Garbage in, garbage out: blockchain cannot verify that the initial data (for example, a sensor reading) was true.
- Energy use in some consensus methods.

- Integration effort with existing ERP and MES.
- Governance: deciding who can write data and who can change rules.

Exam Q: Why does blockchain need a cryptographic hash link between blocks?

6. Smart Contracts and Decentralised Manufacturing Syst

6.1 Smart contracts

A smart contract is a program stored on a blockchain that runs automatically when agreed conditions are met. It removes the need for a human intermediary to enforce the agreement.

- Example: "When the supplier's delivery sensor confirms arrival of 500 kg of resin at the correct temperature, release the payment."
- Written in languages such as Solidity (Ethereum) or chaincode in Hyperledger Fabric.
- Once deployed, the rules are fixed; mistakes in code can be costly, so testing and audits are essential.

6.2 Smart contracts in manufacturing

- Automatic purchase order and payment on delivery.
- Quality-based payment (payment reduced if measured defect rate is high).
- Service-level penalties for late deliveries.
- Automatic reordering when inventory falls below a limit.

6.3 Decentralised manufacturing systems

In a centralised system, one controller or company decides all schedules. In a decentralised system, autonomous units (machines, cells, AGVs, even companies) negotiate and make local decisions based on rules and shared data.

- Advantages: resilience (failure of one unit does not stop the others), flexibility, faster local response, scalability.

- Challenges: achieving a good global result, coordination, trust and standards.

6.4 Example: distributed production scheduling

Each machine agent bids for jobs based on its load and capability. A job is assigned to the agent that offers the best cost and time. If one machine fails, its jobs are re-bid to other agents automatically.

6.5 Decentralised manufacturing in practice

- Distributed additive manufacturing: printing spare parts at many local sites from one verified digital file.
- Blockchain-based machine-to-machine payments for use of shared equipment.
- Supply networks in which suppliers and manufacturers share schedules through a common ledger.

KEY: Decentralised does not mean uncontrolled: rules, identity and standards still govern every participant.

Exam Q: Write one smart contract condition for a supply chain of your choice.

6.6 Blockchain consensus methods (quick notes)

- Proof of Work (PoW): miners solve hard puzzles; secure but energy-heavy; used by Bitcoin.
- Proof of Stake (PoS): validators are chosen according to the stake they lock; far less energy.
- Proof of Authority (PoA) and PBFT: a known set of trusted validators agrees on blocks; fast and suitable for private or consortium chains.

KEY: For factories, permissioned chains with PBFT-style or PoA consensus are usually the practical choice.

2.5 Edge hardware choices

- Programmable logic controllers with built-in communication (fast and certified for industry).
- Industrial PCs running Linux or Windows for data-heavy applications.
- Embedded boards such as single-board computers for simple sensor gateways.
- GPU or AI accelerator modules for running vision and anomaly models near the machine.
- Selection criteria: temperature range, vibration and dust rating, power supply, number of I/O ports, security features and long-term availability of parts.

1.6 Cloud security basics for manufacturing

- Encrypt data in transit (TLS) and at rest.
- Use strong identity management: unique accounts, multi-factor login, role-based access.
- Segment networks: keep the control network separate from the office network and expose only what is needed through a DMZ.
- Monitor logs continuously and keep backups in another region.
- Define who owns the data and where it may be stored (data governance).

KEY: Connecting machines to the cloud increases the attack surface; security must be designed in from the start.

3.6 Steps for rolling out a cloud MES

1. Map current processes: work orders, routing, quality checks and reporting.
2. Define KPIs and the business goal (for example, reduce WIP by 20 percent).
3. Select vendor and check integration with ERP, PLC, SCADA and quality systems.
4. Connect a pilot line through an edge gateway; test data accuracy against manual records.
5. Train operators and supervisors; design simple screens for the shop floor.
6. Go live on the pilot, measure KPIs for 4 to 8 weeks, then extend to other lines.
7. Plan for fallback: if the internet is down, the line should continue with local buffering.

Exam Q: Why is it important to test data accuracy during the pilot phase of an MES project?

5.7 Blockchain versus traditional database

Feature	Traditional database	Blockchain
Control	Single owner or administrator	Shared among participants
Edits	Records can be changed or deleted	Past blocks are protected by hashes
Trust	Trust in the administrator	Trust in the protocol and consensus
Speed	Very fast	Slower, depends on design
Best use	Internal records of one company	Shared records between many companies

- Blockchain is useful when parties do not fully trust one another and need a common record.
- If only one company needs the data, a well-secured database is usually simpler and cheaper.

7. Worked Examples and Revision

7.1 Worked example: RFID-based pallet tracking

- Tag each pallet with a UHF tag holding its unique ID and product code.
- Install readers at receiving, storage, production and dispatch doors.
- Each read is time-stamped and sent through the edge gateway to the WMS (warehouse management system).
- The system shows live stock levels and automatically flags a pallet that leaves the wrong dock.

7.2 Worked example: edge decision for a temperature alarm

- Cold-storage sensor reads 9 degrees Celsius; the limit is 8 degrees.
- The edge gateway triggers a local alarm and starts the backup cooler within 2 seconds.
- A summary is sent to the cloud for records and to the MES for the affected batch.
- Result: the batch is saved without waiting for an internet round trip.

7.3 Revision checklist

- Can you define IaaS, PaaS and SaaS with one example each?
- Can you explain why edge computing reduces latency?
- Can you draw the automation pyramid and place the MES?
- Can you describe the components of an RFID system?
- Can you compare LF, HF and UHF RFID?
- Can you explain blockchain immutability in simple words?
- Can you describe a smart contract with an example?

7.4 Short answer practice

Exam Q: What is the difference between fog and edge computing?

Exam Q: Name two KPIs that an MES tracks and explain one.

Exam Q: Why is a consortium blockchain often preferred over a public one in supply chains?

UNIT IV: Digital Twin, Simulation, AR/VR and Cybersecurity

Subject: 23A03703d - Smart Manufacturing

1. Digital Twin Technology and Virtual Prototyping

1.1 Recap of the digital twin

A digital twin is a living virtual model of a physical product, machine, line or plant. It is updated with real data, so it shows the current state and can predict future behaviour. (Unit II introduced the idea; this unit focuses on design, simulation and security around it.)

KEY: Digital twin = physical asset + virtual model + data link (ideally two-way).

1.2 Digital twin across the product life cycle

- Design: the twin is the CAD and simulation model used to test ideas.
- Build: the twin helps plan the factory layout and test PLC code virtually (virtual commissioning).
- Operate: live data keeps the twin in step with the real asset.
- Maintain: the twin predicts failures and plans service.
- Retire or upgrade: the twin records history, helping redesign the next product.

1.3 Virtual prototyping

Virtual prototyping builds and tests a product on the computer before any physical prototype is made.

- Benefits: fewer physical prototypes, faster design cycles, earlier detection of faults, lower material cost.
- Methods: finite element analysis (FEA) for stress, computational fluid dynamics (CFD) for airflow and cooling, multibody dynamics for motion, thermal analysis for heat.

- Example: a car engine part is tested for stress, temperature and fatigue in simulation before the first casting is poured.

1.4 Virtual commissioning

Virtual commissioning connects the real PLC program to a simulated plant. Errors in logic, interlocks and sequences are found while the machine is still on paper, reducing start-up time on the shop floor.

1.5 Digital twin benefits and difficulties

- Benefits: better decisions, reduced risk, faster time-to-market, knowledge retained in the model.
- Difficulties: building accurate models is expensive; models must be updated; data integration across vendors is hard; skills in simulation and data are scarce.

Exam Q: Explain how virtual commissioning reduces start-up time.

2. Simulation Software for Smart Manufacturing

2.1 Why simulate?

Simulation lets engineers test a system under many conditions without disrupting real production. It answers questions such as: how many machines do we need? where is the bottleneck? what happens if demand doubles?

2.2 Types of simulation

- Discrete event simulation (DES): models events such as job arrivals and machine completions. Best for production lines, queues and logistics.
- Continuous simulation: models quantities that change smoothly, such as temperature, level or chemical concentration.
- Agent-based simulation: models many autonomous agents (machines, AGVs, operators) and their interactions.
- Physics-based simulation: FEA, CFD and multibody models for product and process behaviour.

- Monte Carlo simulation: uses random sampling to study uncertainty, for example variation in cycle times.

2.3 Common simulation tools (examples)

- Plant and logistics: Siemens Plant Simulation, FlexSim, AnyLogic, Arena.
- Physics and engineering: ANSYS, COMSOL, Abaqus, OpenFOAM (CFD).
- Robot simulation: RoboDK, Visual Components, Process Simulate.
- Process simulation: Aspen Plus and HYSYS for chemical plants.
- Open-source options: Python libraries such as SimPy for discrete event simulation.

2.4 Steps in a simulation study

1. Define the objective and the questions to answer.
2. Collect input data (cycle times, failure rates, arrival patterns).
3. Build the model at the right level of detail.
4. Verify the model (does it do what we intended?).
5. Validate the model (does it match real behaviour?).
6. Run experiments and compare scenarios.
7. Analyse results and communicate recommendations.

KEY: A simulation is only as good as its data and its validation; a fancy animation proves nothing on its own.

2.5 Example: line balancing

A line has five stations with cycle times 40, 55, 52, 38 and 60 seconds. A discrete event simulation shows that station 5 limits the output to one unit every 60 seconds. Moving one task from station 5 to station 4 balances the line and raises output without new machines.

Exam Q: What is the difference between verification and validation of a simulation model?

3. Augmented Reality (AR) and Virtual Reality (VR) in Indu

3.1 Definitions

- Virtual reality (VR): a fully computer-generated environment. The user wears a headset and is immersed in a virtual world.
- Augmented reality (AR): digital information is overlaid on the real world, usually through a headset, tablet or smartphone camera.
- Mixed reality (MR): real and virtual objects interact in real time; the virtual objects respond to the real environment.

3.2 AR and VR in manufacturing

- Training: operators practise machine setup or safety procedures in VR before touching real equipment.
- Maintenance: AR glasses show step-by-step repair instructions and wiring diagrams over the machine.
- Remote assistance: an expert sees what the technician sees and draws annotations on the live view.
- Assembly guidance: AR shows which bolt to tighten next and torque values, reducing errors.
- Design review: VR walk-throughs of a factory layout before construction.
- Quality inspection: AR overlays tolerance limits on the part during inspection.

3.3 Hardware and software

- VR headsets: fully immersive displays with tracking of head and hand movement.
- AR devices: smart glasses, tablets, head-mounted displays with cameras and sensors.
- Software: 3D engines such as Unity and Unreal Engine; CAD integration; tracking libraries; IoT data feeds.

3.4 Benefits and challenges

- Benefits: lower training cost, fewer errors, faster repairs, safer practice, knowledge capture from experts.

- Challenges: hardware cost, comfort and eye strain, need for accurate 3D models, network needs for remote help, and the time needed to create content.

KEY: AR and VR are most valuable where the cost of a mistake on the real machine is high.

3.5 Example

A technician fixing a compressor sees the wiring path highlighted on the machine through AR glasses. A remote expert places an arrow on the live view to show a valve. The repair takes 40 minutes instead of two hours spent searching manuals.

Exam Q: Give one example each of AR and VR use in a factory.

4. Cybersecurity in Smart Manufacturing: Threats and Ris

4.1 Why security matters now

Traditional factories were isolated: machines were not connected to the internet. Smart factories connect OT and IT, which opens many new paths for attackers. A single infected laptop can spread malware to controllers that have no protection.

4.2 Types of threats

- Malware and ransomware: encrypt files or stop production until a ransom is paid.
- Phishing and social engineering: staff tricked into giving passwords or opening malicious files.
- Insider threats: employees or contractors who misuse access, intentionally or by mistake.
- Denial of service (DoS): flooding networks so that controllers cannot communicate.
- Man-in-the-middle attacks: intercepting and changing messages between devices.

- Unauthorised access: default passwords on PLCs and HMI panels left unchanged.
- Supply chain attacks: malicious code inserted into software or firmware updates.
- Physical attacks: tampering with USB ports or sensors on the shop floor.

4.3 Risks in smart manufacturing

- Safety risk: manipulated machine settings can injure people or damage equipment.
- Production risk: downtime, lost orders and missed deliveries.
- Quality risk: subtle changes to process data can produce defective products without alarms.
- Intellectual property risk: theft of designs and recipes.
- Financial and legal risk: penalties, insurance claims, regulatory breaches.
- Reputational risk: loss of customer trust.

4.4 Why OT security is different from IT security

Aspect	IT systems	OT systems
Priority	Confidentiality	Availability and safety first
Patching	Frequent and quick	Limited by downtime and certification
Lifetime	3 to 5 years	15 to 30 years
Protocols	Standard internet protocols	Old industrial protocols with no security
Response	Shut down and reboot	Shutdown may be dangerous

KEY: In OT, the rule is "safety, then availability, then integrity, then confidentiality".

4.5 Attack example

Stuxnet (discovered in 2010) is a well-known industrial malware that targeted PLCs. It showed that attackers can reach control systems through IT networks and USB devices. Lessons: segment networks, control removable media and monitor controller behaviour.

Exam Q: Why is patching industrial systems harder than patching office computers?

5. Cybersecurity Frameworks for Industrial Systems

5.1 Purpose of a framework

A framework gives a structured way to identify assets, assess risks, select controls, respond to incidents and improve over time.

5.2 IEC 62443

- International series of standards for security of industrial automation and control systems (IACS).
- Introduces security levels (SL 1 to SL 4) and the concept of zones and conduits: groups of assets with similar risk, and the controlled communication paths between them.
- Covers the whole life cycle: product developers, system integrators and asset owners each have requirements.

5.3 NIST Cybersecurity Framework (CSF)

- Developed by the US National Institute of Standards and Technology. Version 2.0 (2024) adds the govern function.
- Six core functions: Govern, Identify, Protect, Detect, Respond and Recover.
- Widely used because it is flexible and technology-neutral; it can be applied to plants and suppliers.

5.4 ISO/IEC 27001 and 27019

- ISO/IEC 27001 defines an information security management system (ISMS).

- ISO/IEC 27019 gives guidance for the energy sector control systems.

5.5 Defence in depth

Security is built in layers so that a single failure does not expose everything:

- Physical security: locked cabinets, controlled access to the shop floor.
- Network segmentation: firewalls and DMZ between OT and IT.
- Host security: antivirus or allow-listing on engineering workstations.
- Application security: secure coding and signed updates.
- Monitoring: intrusion detection for industrial protocols.
- People and process: training, incident response plans, backups.

KEY: No framework replaces a risk assessment; the framework tells you what to check, and the risk assessment tells you what matters most.

Exam Q: What are the six core functions of the NIST CSF?

6. Best Practices for Securing IoT and IIoT Devices

6.1 Secure design and procurement

- Choose devices with secure boot, signed firmware and encrypted communication.
- Ask suppliers for a software bill of materials (SBOM).
- Avoid devices with hard-coded default passwords.
- Require support and security update commitments for the device lifetime.

6.2 Identity and access

- Give each device a unique identity and certificate.
- Use strong, unique passwords and change defaults on installation.
- Apply the principle of least privilege: each user and device gets only what it needs.
- Use multi-factor authentication for remote access.

6.3 Network security

- Segment networks into zones; limit traffic between zones.
- Disable unused ports and services.
- Use VPNs or secure gateways for remote access; never expose PLCs directly to the internet.
- Monitor network traffic and watch for unusual industrial protocol commands.

6.4 Data protection

- Encrypt data in transit with TLS and, where possible, OPC-UA security.
- Protect data at rest with encryption and access control.
- Keep backups offline or immutable, and test restores regularly.

6.5 Maintenance and monitoring

- Maintain an inventory of all connected devices.
- Patch on a tested schedule; use compensating controls when patching is not possible.
- Log events centrally and review logs regularly.
- Run vulnerability scans and penetration tests with care to avoid disrupting production.

6.6 People and response

- Train operators and engineers to recognise phishing and unusual machine behaviour.
- Prepare an incident response plan with clear roles and communication steps.
- Practise drills so that the team knows what to shut down, what to isolate and whom to call.
- Share threat intelligence with industry groups and computer emergency response teams.

6.7 Checklist for an IIoT deployment

- Inventory of devices and their owners.
- Risk assessment and zone design.
- Secure configuration and identity for each device.

- Encrypted communication with certificates.
- Monitoring, logging and alerting in place.
- Backup, recovery and response plan tested.

Exam Q: List three best practices for securing IIoT sensors.

1.6 Digital twin architecture for manufacturing (ISO 2324

ISO 23247 is an international framework for digital twins in manufacturing. It describes four main entities:

- Observable manufacturing elements (OMEs): the real things such as machines, tools, materials and people.
- Data collection and device control: sensors and controllers that gather and act on data.
- Digital twin entity: the virtual model with its data, logic and visualisation.
- User application: the people and systems that use the twin, for example planners or maintenance teams.

KEY: Keep the framework in mind: real elements feed data into a twin, and users act on the twin's results.

2.6 Simulation and digital twins together

- A simulation is often a one-time study; a digital twin keeps running with live data.
- Simulation models can be used inside a twin as its "what-if" engine.
- Example: a twin of a cooling system runs a physics model every minute with live temperature data; the operator tests a fan-speed change in the twin before applying it to the real plant.

4.6 Security tools commonly used in OT environments

- Firewalls designed for industrial protocols (Modbus, OPC-UA, PROFINET) that filter by function code.
- Industrial intrusion detection systems that learn normal communication patterns.
- Asset inventory tools that discover devices on the network and list firmware versions.
- Application allow-listing on engineering workstations so that only approved programs run.
- Data diodes: one-way hardware links that let data leave the OT network but not enter it.
- Secure remote access gateways with session recording.

5.6 Case study: securing a bottling plant

A bottling plant connected its PLCs to a cloud analytics platform to reduce waste. Early reviews found several issues: default passwords on HMI panels, a shared engineering laptop used for internet browsing, and open ports on the PLC network.

- Actions taken: passwords changed and unique accounts created; laptop separated into an engineering-only machine; firewall rules allowed only the required OPC-UA traffic; a data diode was added for data flowing to the cloud.
- Results: analytics continued without loss of function, and the audit found no unauthorised access in the following year.
- Lesson: most early wins come from basic hygiene, not from expensive tools.

6.8 Summary table of best practices

Area	Key practice	Example
Design	Secure by default	Disable unused services
Identity	Unique credentials	Per-user accounts on HMI
Network	Segmentation	Firewall between OT and IT
Data	Encryption	TLS on MQTT and OPC-UA
Operations	Patch and monitor	Scheduled maintenance window
People	Training and drills	Phishing awareness course

7. Worked Examples and Revision

7.1 Worked example: zone design for a small plant

- Zone A: PLCs and drives on the production line. Access only from the engineering workstation.
- Zone B: SCADA and HMI servers. Allowed to talk to Zone A through a firewall with allow-listed Modbus or OPC-UA traffic.
- Zone C: MES and office IT network. Connected to Zone B through a data diode or a tightly controlled gateway.
- Remote vendors connect through a VPN into a jump server in a separate DMZ, with recorded sessions.

7.2 Worked example: responding to ransomware on an HMI

1. Detect: monitoring alerts on unusual file encryption.
2. Contain: isolate the infected HMI from the network; keep the line in a safe state.
3. Eradicate: rebuild the HMI from a known-good image; check PLC programs against backups.
4. Recover: restore data, test functions, then restart production in stages.

5. Learn: update the zone rules and training.

7.3 Revision checklist

- Can you explain how digital twins support virtual commissioning?
- Can you name three simulation types and their uses?
- Can you distinguish AR, VR and MR?
- Can you list four threats to a smart factory?
- Can you explain why OT security puts safety first?
- Can you name the six NIST CSF functions?
- Can you describe zones and conduits in IEC 62443?

7.4 Short answer practice

Exam Q: Differentiate between verification and validation in simulation.

Exam Q: Why should PLCs never be directly exposed to the internet?

Exam Q: Explain defence in depth with three layers.

UNIT V: Sustainable Manufacturing, Case Studies and Future Trends

Subject: 23A03703d - Smart Manufacturing

1. Sustainable and Green Manufacturing

1.1 Definition

Sustainable manufacturing creates products through processes that minimise negative environmental impact, conserve energy and natural resources, are safe for employees and communities, and are economically viable. Green manufacturing is the environmental part of this idea: less pollution, less waste, less energy and fewer harmful substances.

KEY: Sustainability has three pillars: environment (planet), economy (profit) and society (people). A good smart factory improves all three.

1.2 Why it matters

- Regulation: carbon taxes, emission limits and reporting rules (for example, the EU Carbon Border Adjustment Mechanism and corporate disclosure rules).
- Customers: buyers ask for low-carbon products and transparent supply chains.
- Costs: energy and material prices are volatile; efficiency protects margins.
- Risk: climate events and resource shortages disrupt production.

1.3 The 6 R's of green manufacturing

- Reduce: use fewer materials and less energy.
- Reuse: use parts or containers more than once.
- Recycle: turn scrap back into material.
- Recover: extract energy or materials from waste.
- Redesign: design products for disassembly and longer life.

- Remanufacture: restore used products to like-new condition.

1.4 Life cycle assessment (LCA)

LCA measures the environmental impact of a product from raw material to disposal. It counts energy, emissions, water and waste at every stage. It helps decide where improvement matters most.

- Cradle-to-gate: from raw material to the factory gate.
- Cradle-to-grave: the full life cycle including use and disposal.

2. Energy Efficiency in Smart Manufacturing

2.1 Where energy is used

In many plants, electric motors, compressed air, HVAC, lighting, furnaces and ovens consume most of the energy. Compressed air is often a surprisingly expensive utility because leaks waste large amounts of power.

2.2 Measures to improve efficiency

- Energy monitoring: sub-metering each machine line to find waste.
- Variable speed drives: match motor speed to load instead of running at full speed.
- Leak detection for compressed air: ultrasonic detectors find leaks quickly.
- Heat recovery: capture waste heat from furnaces and compressors.
- LED lighting with occupancy and daylight sensors.
- Power factor correction to reduce reactive losses and penalties.
- Load shifting: run energy-intensive processes when electricity is cheaper or greener.

2.3 Energy KPIs

- Energy per unit produced (kWh per part).
- Energy intensity (energy per unit of revenue or output).
- Carbon intensity (kg CO_2 per unit).
- Specific energy consumption for processes such as melting or drying.

KEY: You cannot improve what you do not measure: start with energy metering and a baseline.

2.4 Example: motor optimisation

A plant has 40 motors running at full speed. Adding variable speed drives to 15 fan and pump motors reduces their average consumption by 30 percent because power falls steeply with speed for fans and pumps. The payback period is 2 years, and the energy saving continues for the drive life.

2.5 Role of smart systems

- Digital energy management systems (EMS) collect meter data and show it on dashboards.
- Machine learning finds unexpected consumption and suggests setpoint changes.
- Digital twins test energy-saving changes before applying them.

Exam Q: Name three ways to reduce energy use in a compressed air system.

3. Sustainable Supply Chain Management

3.1 What it means

Sustainable supply chain management (SSCM) considers environmental and social impacts across all suppliers, transport and logistics, not only inside the factory. Many emissions and risks sit in the supply chain, often more than half of a product's footprint.

3.2 Key practices

- Supplier selection with sustainability criteria and audits.
- Low-carbon transport: optimised routes, consolidated loads, electric vehicles, rail where possible.
- Local sourcing where practical to cut transport emissions.
- Packaging reduction and returnable containers.
- Reverse logistics: taking back used products for reuse and recycling.

- Responsible sourcing of minerals and materials, with traceability.

3.3 Technology support

- IoT tracking of shipments and temperature for food and pharmaceuticals.
- Blockchain traceability for certifications and origin claims.
- AI route optimisation to reduce kilometres and fuel.
- Digital product passports that store material composition and recycling instructions.

3.4 Circular supply chains

A circular supply chain keeps materials in use for as long as possible. Products are designed for return, parts are remanufactured, and secondary materials replace virgin materials.

KEY: A circular supply chain needs data about the product: what it is made of, where it is, and when it comes back.

3.5 Example

A furniture maker tags each product with a QR code linking to a digital product passport. When a customer returns a sofa, the repair team knows the frame and fabric type, orders the right spare part, and refurbishes the item for resale.

Exam Q: Explain why supply chain emissions are important even for a factory with clean operations.

4. Role of AI in Green Manufacturing

4.1 AI for energy optimisation

- Predicts energy demand and suggests load schedules.
- Optimises setpoints for HVAC, compressors and furnaces.
- Detects unusual consumption that could mean leaks or faults.

4.2 AI for waste and material efficiency

- Vision systems detect defects early, reducing scrap and rework.

- Models optimise cutting patterns to reduce material offcuts.
- Process models find the best temperature and time settings for lower energy per part.

4.3 AI for product design and life cycle

- Generative design creates lighter parts with the same strength, using less material.
- AI estimates the environmental impact of design options during early design.
- Predictive maintenance extends equipment life and avoids wasteful breakdowns.

4.4 AI for supply chain sustainability

- Demand forecasting reduces overproduction and excess inventory.
- Route optimisation lowers transport emissions.
- Supplier risk models flag high-impact suppliers for audit.

4.5 Cautions

- AI systems use energy; the benefit must exceed the cost of computing.
- Models must be trained on good data and monitored for bias and drift.
- Human judgement remains important, especially for safety and ethics.

KEY: The best green AI project is one that saves more energy and material than it uses.

Exam Q: Give one example of how AI can reduce material waste in manufacturing.

5. Case Studies of Leading Smart Manufacturing Compani

5.1 Siemens: digitalised factories

Siemens operates digitalised factories, including the Amberg electronics plant in Germany. The company uses its digital twin software, PLC automation and MES to plan and operate lines. Its experience is often shared with customers as a reference model: integrated engineering, production and data.

- Lesson: connect engineering (design) and operations (production) to shorten change cycles.

5.2 Tesla: gigafactories and automation

Tesla's factories use large numbers of robots and an integrated approach to design and manufacturing, with many processes designed for high volume. Large castings and simplified designs reduce the number of parts. The company is known for rapid design changes in production.

- Lesson: design for manufacturing and automation from the start.
- Note: claims about production methods vary; verify with official reports before relying on specific figures.

5.3 Toyota: lean and smart production

Toyota's production system (lean manufacturing) is based on just-in-time delivery, jidoka (automation with a human touch: machines stop when a problem appears), and continuous improvement (kaizen). Smart technologies are being added to support these principles.

- Lesson: digital tools work best when they support sound process discipline.

5.4 Bosch: connected manufacturing

Bosch has built connected plants that use IoT sensors, edge computing and analytics to improve quality and reduce downtime. It also develops industrial IoT platforms for other manufacturers.

- Lesson: share knowledge between plants so that improvements scale quickly.

5.5 Comparing approaches

Company	Main focus	Key technology	Core lesson

Siemens	Integrated digital factory	Digital twin, PLC, MES	Link engineering and operations
Tesla	High-volume automation	Robots, design simplification	Design for manufacturing
Toyota	Lean production	Jidoka, JIT, kaizen	Process discipline first
Bosch	Connected plants	IoT, edge, analytics	Scale improvements across plants

Exam Q: Choose one case study and explain its lesson in your own words.

6. Emerging Technologies in Smart Manufacturing

6.1 5G and private wireless networks

5G offers low latency, high reliability and support for many devices. Private 5G networks inside a plant can connect mobile robots, AR devices and sensors with predictable performance.

6.2 Time-Sensitive Networking (TSN)

TSN adds deterministic timing to standard Ethernet. Control, vision and motion traffic can share one network without losing timing guarantees.

6.3 Generative AI and large language models

- Help engineers write PLC code, documentation and maintenance instructions.
- Let operators ask questions of machine manuals and historical records in plain language.
- Must be used with safeguards, because generated code and advice need verification before use on machines.

6.4 Autonomous systems and swarm robotics

Groups of mobile robots coordinate to move materials, inspect areas and assist in assembly without central control, improving flexibility and resilience.

6.5 Advanced materials and new manufacturing processes

- Metal additive manufacturing for lightweight and complex parts.
- Smart materials that change shape or report stress.
- Micro and nano manufacturing for electronics and medical devices.

6.6 Quantum computing (future outlook)

Quantum computers may in future solve certain optimisation problems, such as complex scheduling or materials simulation, much faster than classical computers. Practical industrial use is still at an early stage.

6.7 Edge AI and TinyML

Very small machine learning models run on low-power microcontrollers at sensors. This allows detection of vibration or sound patterns without any cloud connection.

KEY: Emerging technologies should be adopted according to a clear value case, not because they are new.

7. Future of Industry 5.0 and Beyond

7.1 What is Industry 5.0?

Industry 5.0 is a vision promoted by the European Union (around 2021) that complements Industry 4.0. It places the worker at the centre of production and adds sustainability and resilience as explicit goals.

- Human-centric: technology supports people, with cobots as partners rather than replacements; focus on skills, wellbeing and inclusion.
- Sustainable: energy and resource use are designed in from the start.
- Resilient: factories can keep operating during crises, such as supply shortages or disasters.

7.2 Differences between Industry 4.0 and 5.0

Aspect	Industry 4.0	Industry 5.0
Main focus	Efficiency and connectivity	Human values, sustainability and resilience
Role of people	Operators supervise machines	People and machines collaborate
Key technologies	IoT, cloud, AI, CPS	Adds human-machine interfaces, ethics and green tech
Goal	Productivity	Productivity with purpose

7.3 Skills for the future workforce

- Data literacy: understanding dashboards and models.
- Digital skills: programming, network and cybersecurity awareness.
- Systems thinking: seeing how processes, people and data interact.
- Adaptability and continuous learning.
- Problem solving and teamwork across engineering and business.

7.4 Future trends beyond Industry 5.0

- Fully autonomous, self-optimising factories with human supervision.
- Circular economy as the default business model.
- Digital twins of entire supply networks.
- Personalised products produced on demand at local micro-factories.
- Ethical AI and transparent decision systems on the shop floor.

7.5 Challenges ahead

- Cost and skills gap, especially for small and medium enterprises.
- Standards and interoperability across vendors.
- Cybersecurity as connectivity grows.
- Balancing automation with job quality.
- Climate targets that require deep reductions in energy and material use.

Exam Q: What does "human-centric" mean in Industry 5.0?

2.6 Carbon accounting: Scope 1, 2 and 3

Companies report greenhouse gas emissions in three scopes:

- Scope 1: direct emissions from sources the company owns, such as furnaces, boilers and company vehicles.
- Scope 2: indirect emissions from purchased electricity, heat and steam.
- Scope 3: all other indirect emissions across the value chain, such as purchased materials, transport, product use and end-of-life disposal.

Scope	Example in a factory	Who controls it
1	Gas burned in a furnace	The plant
2	Electricity bought from the grid	Energy supplier mix and plant efficiency
3	Steel purchased from a supplier	Suppliers and design choices

KEY: Scope 3 is often the largest share, which is why supply chain engagement is essential.

6.8 Green production technologies

- Cleaner production: changing the process to avoid pollution at source, for example replacing a toxic solvent with water-based cleaning.
- Water recycling: treating and reusing process water in cooling and washing, reducing intake and discharge.
- Zero liquid discharge (ZLD): systems that recover nearly all water from effluent so that only solids are removed.
- Dry machining and minimum quantity lubrication (MQL): less cutting fluid to dispose of and clean.
- Heat pumps and waste heat recovery for drying and heating processes.

- Renewable energy: on-site solar panels and power purchase agreements for wind or solar electricity.
- Bio-based and recycled materials for packaging and components.

5.6 Lessons from green case practice

- Start with a baseline: measure energy, water, waste and emissions before setting targets.
- Set specific targets such as reducing scrap by 15 percent or cutting energy per unit by 10 percent in two years.
- Engage employees: ideas from operators often produce the best low-cost improvements.
- Report transparently using recognised standards such as GRI or CDP.
- Link sustainability KPIs to the same dashboards used for productivity, so they are managed together.

7.6 Ethics and human-centric design

- Transparency: workers should know what data is collected about them and how it is used.
- Fairness: AI decisions on scheduling or performance must be checked for bias.
- Safety: cobots and autonomous vehicles must be designed around human behaviour and error.
- Inclusion: interfaces should suit workers of different ages, languages and abilities.
- Participation: workers should help design new workstations and processes.

- Job transition: retraining and redeployment programmes should accompany automation projects.

KEY: Industry 5.0 asks not only "can we automate this?" but also "should we, and how will it affect people?"

7.7 Resilience in practice

- Dual sourcing of critical parts to avoid single-supplier failure.
- Digital twins of supply networks to test disruption scenarios.
- Buffer stock for high-risk items and flexible production lines that can switch products.
- Energy backup and microgrids to keep critical loads running during outages.
- Regular risk drills that combine IT, operations and procurement.

8. Worked Examples and Revision

8.1 Worked example: green business case

A plant wants to cut its compressed air cost. Current cost: 120,000 per year. Survey finds 25 percent leakage. Fixing leaks costs 15,000 and saves about 25 percent of the cost, which is 30,000 per year.

- Simple payback = 15,000 / 30,000 = 0.5 year (six months).
- Add a variable speed compressor later for further savings.
- Result: a quick, strong business case that also reduces emissions.

8.2 Worked example: circular packaging

A company replaces single-use cardboard boxes with a returnable plastic crate used 40 times. If each crate replaces 40 boxes and each box costs 1.5 to produce and ship, the savings per crate cycle are visible in the cost and the waste reduction. Tracking each crate with RFID shows where crates are lost and how to reduce loss.

8.3 Revision checklist

- Can you define sustainable and green manufacturing?
- Can you list the 6 R's?
- Can you explain life cycle assessment in one sentence?
- Can you give three ways to reduce energy in a plant?
- Can you describe one AI application for green manufacturing?
- Can you summarise one case study and its lesson?
- Can you compare Industry 4.0 and Industry 5.0?

8.4 Short answer practice

Exam Q: Why is supply chain sustainability important for a manufacturer?

Exam Q: What is the difference between recycling and remanufacturing?

Exam Q: Name two emerging technologies and one possible use for each in a factory.